

Ke GUO

E-mail: guoke9612@gmail.com * *Phone:* +86 13653629612

Linkedin: KeGUO * *Github:* GUOkekklk * *Homepage:* Ke Nash GUO

Education

Master Degree

ADVANCED ROBOTICS

GPA:3.54/4 **Rank:**1/27

Main courses: Advanced Modeling of Robots; Artificial Intelligence; Computer Vision; Advanced Visual Geometry; Task-based Control; Autonomous Vehicle

Ecole Centrale de Nantes, France

09/2021 - 04/2024

Exchange Program

SIGNAL, CONTROL AND ROBOTICS

GPA:16.9/20

Main courses: Control Systems; Embedded Systems; Basics of Signal Processing and Image Methodology

Ecole Centrale de Nantes, France

10/2020 - 05/2021

Bachelor Degree

MATHEMATICS (IN INFORMATION AND COMPUTING SCIENCE TRACK)

GPA:82.37/100; **Rank:**12/59

Main courses: Mathematical Analysis; Higher Algebra; Probability and Statistics; Numerical Analysis; Foundations of Big Data Analysis; Mathematical Theory of Complex Systems; Ordinary Differential Equation; Numerical Methods for Partial Differential Equations; Operational Research

Harbin Institute of Technology at Weihai, China

09/2017 - 06/2021

Technical Skills

Programming Languages

Python, C#, C++, TypeScript, SQL

AI & Computer Vision

PyTorch, PyTorch Lightning, OpenCV, WandB, Transformers

Robotics & Simulation

ROS, NVIDIA Isaac Sim, Isaac Lab, Isaac Sim Replicator

Backend & Data

ASP.NET Core, FastAPI, NATS, PostgreSQL, TimescaleDB

Frontend Development

React, React Flow, Tailwind CSS

LLM & RAG

Dify, LangChain, Unstructured

Development & DevOps

Linux, Docker, Nginx, Git, Azure CI/CD

Work Experience

Robot Software Engineer (Full-Stack)

01/2025 – Present

ABB Robotics, Shanghai, China

- Design and develop full-stack IoT and predictive maintenance solutions using Python, C#, and TypeScript on a Docker-based microservices architecture.
- Spearhead projects at the intersection of AI-powered productivity, Physical AI, and robotics, applying technologies such as NVIDIA Isaac Lab and LangChain to industrial use cases.
- Bridge software engineering, artificial intelligence, and industrial automation to deliver scalable intelligent systems with measurable business value.
- Mentor interns throughout project delivery and organize internal technical talks on emerging AI technologies.

Trainee

03/2024 – 01/2025

ABB Robotics, Shanghai, China

- Completed rotational training across the painting and testing departments, gaining hands-on experience in robot commissioning, firmware testing, and system validation.
- Collaborated with global teams and developed a cross-functional understanding of industrial robotics.

Research Engineer Intern

02/2023 – 07/2023

Visual Behavior, Lyon, France

- Developed deep learning applications for visual SLAM (Simultaneous Localization and Mapping), including a Transformer-based approach to replace the conventional PnP (Perspective-n-Point) algorithm.
- Researched autonomous positioning and front-end visual odometry based on BEV (Bird's-Eye View) representations.
- Integrated SuperPoint and LightGlue to improve the robustness of feature detection and matching.
- Conducted the research under the supervision of **Thibault Neveu**.

Projects

IoT-Based Predictive Maintenance Platform for Spray-Painting Robots 01/2025 – Present

ABB Robotics, Shanghai, China

- Develop and maintain a full-stack industrial IoT platform for data acquisition, real-time monitoring, and predictive maintenance using a Docker-based microservices architecture.
- Build high-throughput data ingestion pipelines with NATS to process telemetry from robots, PLCs using OPC UA, and IPS (Integrated Painting System).
- Design a pluggable algorithm pipeline using decorators to support time-series processing, data analysis, and anomaly detection.
- Develop an interactive drag-and-drop interface for visual algorithm orchestration using React, React Flow, and Tailwind CSS.
- Maintain an Azure DevOps CI/CD pipeline that automates the building, testing, and deployment of software packages across development and production environments.
- **Tech Stack:** TypeScript, C#, Python, React, ASP.NET Core, FastAPI, NATS, PostgreSQL, TimescaleDB, Docker, Nginx, Azure DevOps.

Vision-Language-Action Model Training with IsaacLab Synthetic Data 01/2026 – 05/2026

ABB Robotics, Shanghai, China

- Developed realistic robotic simulation environments in NVIDIA IsaacLab for collecting manipulation demonstrations.
- Generated LeRobot datasets containing multiview observations and robot actions for training VLA (Vision-Language-Action) models such as $\pi_{0.5}$.
- Trained and evaluated VLA models in simulation before deploying them to physical ABB industrial robots.
- **Tech Stack:** Python, LeRobot, NVIDIA Isaac Lab, VLA.

MCP-Based Robot Control via ABB RWS

10/2025 – 11/2025

ABB Robotics, Shanghai, China

- Developed an MCP (Model Context Protocol) server that exposes ABB RWS (Robot Web Services) capabilities as standardized tools for AI agents.
- **Tech Stack:** Python, MCP, ABB RWS, REST API.

RAG-Based Internal Document Question-Answering System

06/2025 – 08/2025

ABB Robotics, Shanghai, China

- Developed a compliance-aware question-answering system for internal documents using Dify and RAG (Retrieval-Augmented Generation).
- Built a document ingestion pipeline using Unstructured to parse and chunk heterogeneous documents, improving content retrieval for downstream LLM (Large Language Model) generation.

- Implemented request auditing and structured query transformation before LLM inference, enabling controlled access to internal document repositories through the Azure Playground API.
- **Tech Stack:** Python, Dify, LangChain, Azure, Unstructured.

Synthetic Image Generation with IsaacSim and Replicator
01/2025 – 04/2025

ABB Robotics, Shanghai, China

- Built synthetic image-generation pipelines using NVIDIA Isaac Sim and Isaac Replicator for robotic perception tasks.
- Configured virtual environments, cameras, lighting, and object properties to generate annotated image datasets for downstream vision model training.
- **Tech Stack:** Python, NVIDIA Isaac Sim, Isaac Replicator, Computer Vision.

Transformer-Based Visual SLAM
02/2023 – 07/2023

Visual Behavior, Lyon, France

- Designed a Transformer-based model to replace the conventional PnP (Perspective-n-Point) algorithm in visual SLAM (Simultaneous Localization and Mapping), directly estimating camera poses from raw images.
- Integrated SuperPoint and LightGlue to enable robust feature detection and matching for a BEV (Bird's-Eye-View) visual odometry front end.
- Developed and trained deep learning pipelines using PyTorch Lightning.
- **Tech Stack:** Python, PyTorch Lightning, Transformers, SuperPoint, LightGlue.

Honors and Awards

4th Place, NVIDIA Spark Hackathon	04/2026
IP Star Award, ABB Robotics, China	12/2025
Outstanding Trainee, ABB Robotics, China	01/2025
Centrale Nantes Elite Scholarship (6000€), France	06/2021
People's Scholarship (Top 15%) in HIT, China	06/2020
Second prize, National College Student Mathematics Competition, China	12/2019
Second prize, National College Students Mathematical Contest in Modeling, China	08/2018

Language Skills

Chinese	Native
English	Full professional proficiency
French	Limited working proficiency

Research Interest

Physical AI, Robot Perception & Control, VLA, Industrial IoT, Human-Robot Collaboration, Visual SLAM

Others

Interests	Music, Basketball, Fitness
Activities	Active participant and organizer of company events; continuous follower and sharer of the Physical AI ecosystem
Mentorship & Sharing	Supervised interns on an annual basis; regularly organized internal technical sharing sessions on emerging technologies and AI topics